Professional, Legal and Ethical Issues

1. Data Protection and Learners' Personal Information

The system stores personal information such as learners' names, email addresses, user accounts, department details and enrolment records. This information must be collected and processed responsibly.

In Lesotho, the Data Protection Act, 2011 provides a legal framework for protecting personal information. The system should follow its principles by ensuring that personal information is collected for legitimate academic purposes, kept accurate, protected against unauthorized access and not retained longer than necessary.

The design addresses these concerns through:

- Authentication: Users must log in before accessing the system.
- Role-based access control: Students, Lecturers, Heads of Department and Administrators have different permissions.
- Password protection: Passwords are hashed using libsodium rather than stored as plain text.
- Limited access: Students can access their own academic information, while lecturers and Heads of Department access records according to their responsibilities.
- Purpose limitation: Personal information is used to support academic activities such as enrolment, assignment submission, grading and moderation.

The institution should also inform learners about what information is collected, why it is collected, who can access it and how long it is retained.

2. Security of Assessment Records

Assessment records include assignment submissions, marks, lecturer feedback, moderation decisions and released results. These records must be protected against unauthorized viewing, modification, deletion or disclosure.

If assessment records are changed without permission, learners may receive incorrect results or be treated unfairly. The system therefore uses several security measures:

- User authentication prevents unauthenticated users from accessing protected functions.
- Role-based permissions restrict grading to lecturers and moderation and result release to authorized Heads of Department.

- Password hashing protects stored passwords if the database is accessed by an unauthorized person.
- Moderation workflow requires marks to be reviewed before they are released to students.
- Audit logging records mark changes and the reasons for those changes, supporting accountability.
- Database constraints and validation help maintain the integrity and consistency of stored records.

The institution should additionally protect database files through secure storage, restricted file-system permissions and regular backups. Access to the database file itself must also be controlled because application-level login restrictions alone cannot prevent someone with direct file access from changing SQLite records.

3. Accessibility for Users with Disabilities

The LMS should be usable by learners and staff with different abilities, including people with visual, hearing, motor or cognitive disabilities. An inaccessible system could prevent some learners from participating equally in academic activities.

The current system is a console-based C++ application. Although it provides core academic functionality, accessibility should be considered in its interface and future development.

The design should support:

- Clear and consistent menus with descriptive text.
- Keyboard navigation without requiring a mouse.
- Readable text and sufficient contrast.
- Clear error messages and instructions.
- Avoiding reliance on colour alone to communicate important information.
- Compatibility with assistive technologies, where possible.
- Accessible formats for assignment instructions and submitted learning materials.

These measures would make the system easier to use for a wider range of users. However, the current implementation should be tested with users with disabilities and assistive technologies before it can be claimed to be fully accessible.

4. Fairness of the Late-Penalty Rules

The LMS applies a graduated late-penalty policy to encourage learners to submit assignments on time while maintaining consistent treatment.

The current policy is:

Submission time	**Penalty**
On or before the deadline	0%
1 day late	10% of maximum mark
2 days late	20% of maximum mark
3 days late	30% of maximum mark
More than 3 days late	Final mark is 0

Although applying the same rules to all learners promotes consistency, strict penalties may disadvantage students who experience circumstances beyond their control, such as illness, disability-related difficulties, family emergencies or technical problems.

The system addresses fairness by:

- Applying the same penalty policy consistently.
- Accepting late submissions rather than automatically rejecting all late work.
- Allowing lecturers to grant individual extensions where justified.
- Recording the revised deadline and reason for an extension.
- Applying penalties based on the applicable deadline, including an approved extension.
- Allowing academic staff to review submissions and marks through the grading and moderation process.

The institution should clearly communicate the penalty policy to learners before assignments are due. It should also establish transparent procedures for requesting extensions and appealing decisions. This helps ensure that the policy is applied consistently without ignoring legitimate exceptional circumstances.

5. Professional Responsibilities and Accountability

The people responsible for operating and maintaining the LMS have a professional duty to protect learners and maintain the reliability of academic information.

The design supports these responsibilities by separating the duties of Administrators, Lecturers and Heads of Department. For example, lecturers award marks, Heads of Department moderate

and release results, and students view their own released results. This separation reduces the risk of misuse of authority and helps ensure that academic decisions can be traced.

Staff should also receive training on confidentiality, responsible system use and the correct handling of learner information. Access permissions should be reviewed regularly, particularly when staff change roles or leave the institution.

Conclusion

The LMS addresses important professional, legal and ethical concerns through authentication, role-based access, password hashing, controlled grading and moderation, audit records and extension procedures. However, these features are only part of responsible system operation. The institution must also implement secure database backups, retention and deletion procedures, accessibility testing, staff training and clear policies for handling learner information and exceptional academic circumstances.

Applicable legislation: Lesotho's Data Protection Act, 2011 (Act No. 5 of 2012) is particularly relevant. Sections 15–20 address lawful processing, minimality, purpose limitation, retention and security; sections 24–27 address information quality, notification, access and correction. The institution should ensure that its actual practices comply with these requirements, rather than assuming that the software features alone guarantee compliance.

Legislation and official source

1. Kingdom of Lesotho. (2012). Data Protection Act, 2011 (Act No. 5 of 2012). Government Gazette No. 19, 22 February 2012.

Book

2. Stallings, W. (2018). Effective Cybersecurity: A Guide to Using Best Practices and Standards. Addison-Wesley Professional.

Assessment fairness and academic practice

3. Brown, S., & Knight, P. (1994). Assessing Learners in Higher Education. Kogan Page.